

# Steven D. Morad

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## EDUCATION

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### University of Cambridge 2024

*Ph.D. in Computer Science*

- Advisor: Prof. Amanda Prorok
- Thesis: Deep Memory Models and Efficient Reinforcement Learning under Partial Observability

### University of Arizona 2019

*M.S. in Aerospace Engineering*

- Advisor: Prof. Jekan Thanga
- Thesis: The Spinning Projectile Extreme Environment Robot

### University of California, Santa Cruz 2015

*B.S. in Computer Science (Honors)*

## WORK EXPERIENCE

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### University of Macau 2024 – Present

*Assistant Professor*

Research, teaching, and supervision in deep learning, reinforcement learning, and robotics

### University of Cambridge 2024

*Adjunct Lecturer*

Designed and taught a deep reinforcement learning module

### Anyscale 2022 – 2023

*Contract Software Engineer (Reinforcement Learning)*

Development of RLlib, an open-source reinforcement learning library

### Toshiba Research Europe 2021 – 2022

*Graduate Research Intern*

Vision-only navigation for drones and ground vehicles using reinforcement learning

### NASA Jet Propulsion Lab 2018, 2019

*Graduate Robotics Research Intern*

Visual inertial odometry and robot dynamics for Enceladus and Europa

### Meta 2015 – 2017

*Production Engineer II*

Software development and systems engineering for Meta's distributed compute engine

## SELECTED PUBLICATIONS AND PATENTS

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Investigating Memory in Model-Free RL with POPGym Arcade.

Wang, Z., He, Z., Zhang, B., Toledo, E., **Morad, S.D.**

**(Spotlight, Top 2.2%) International Conference on Machine Learning.**

Explore to Learn: Latent Exploration through Disentangled Synergy Patterns for Reinforcement Learning in Overactuated Control.

Wang, Y., Zhao, K., Li, X., Li, Y., Chen, J., **Morad, S.D.**, Leong, H.

**(Oral, Top 5%) AAAI Conference on Artificial Intelligence.**

Language-Conditioned Offline RL for Multi-Robot Navigation.

**Morad, S.D.**, Shankar, A., Blumenkamp, J., Prorok, A.

**International Conference on Robotics and Automation.**

Recurrent Reinforcement Learning with Memoroids.  
**Morad, S.D.**, Lu, C., Kortvelesy, R., Liwicki, S., Foerster, J., Prorok, A.  
**Neural Information Processing Systems.**

CoViS-Net: A Cooperative Visual Spatial Foundation Model for Multi-Robot Applications.  
 Blumenkamp, J., **Morad, S.D.**, Gielis, J., Prorok, A.  
**Conference on Robot Learning.**

Task Performing Agent Systems and Methods.  
**Morad, S.D.**, Mecca, R., Poudel, R., Liwicki, S., Cipolla, R.  
**US Patent US12085947B2, UK Patent GB2598758B, Japan Patent JP7225292B2.**

Reinforcement Learning with Fast and Forgetful Memory.  
**Morad, S.D.**, Kortvelesy, R., Liwicki, S., Prorok, A.  
**Neural Information Processing Systems.**

Generalised  $f$ -Mean Aggregation for Graph Neural Networks.  
 Kortvelesy, R., **Morad, S.D.**, Prorok, A.  
**Neural Information Processing Systems.**

POPGym: Benchmarking Partially Observable Reinforcement Learning.  
**Morad, S.D.**, Kortvelesy, R., Liwicki, S., Prorok, A.  
**International Conference on Learning Representations.**

Permutation-Invariant Set Autoencoders with Fixed-Size Embeddings for Multi-Agent Learning.  
 Kortvelesy, R., **Morad, S.D.**, Prorok, A.  
**International Conference on Autonomous Agents and Multiagent Systems.**

A Framework for Real-world Multi-robot Systems Running Decentralized GNN-based Policies.  
 Blumenkamp, J., **Morad, S.D.**, Gielis, J., Li, Q., Prorok, A.  
**International Conference on Robotics and Automation.**

Modeling Partially Observable Systems using Graph-Based Memory and Topological Priors.  
**Morad, S.D.**, Liwicki, S., Kortvelesy, R., Mecca, R., Prorok, A.  
**Learning for Dynamics and Control.**

Embodied Visual Navigation with Automatic Curriculum Learning in Real Environments.  
**Morad, S.D.**, Mecca, R., Poudel, R., Liwicki, S., Cipolla, R.  
**Robotics and Automation Letters.**

Improving Visual Feature Extraction in Glacial Environments.  
**Morad, S.D.**, Nash, J., Higa, S., Smith, R., Parness, A., Barnard, K.  
**Robotics and Automation Letters.**

A Spring Propelled Extreme Environment Robot for Off-world Cave Exploration.  
**Morad, S.D.**, Dailey, T., Vance, L.D., Thangavelautham, J.  
**IEEE Aerospace Conference.**

## AWARDS, FELLOWSHIPS & GRANTS

|                                                                     |                             |
|---------------------------------------------------------------------|-----------------------------|
| <b>Funding Scheme for Research and Innovation (Co-Investigator)</b> | <b>2025</b>                 |
| <i>Macau FDCT</i>                                                   | <i>MOP 1.9M (GBP ~190k)</i> |
| <b>Enhancement Programme</b>                                        | <b>2025</b>                 |
| <i>University of Macau</i>                                          | <i>MOP 180k (GBP ~18k)</i>  |
| <b>Startup Research Grant</b>                                       | <b>2025</b>                 |
| <i>University of Macau</i>                                          | <i>MOP 150k (GBP ~15k)</i>  |
| <b>2nd Best Research Talk</b>                                       | <b>2023</b>                 |
| <i>Jesus College Graduate Conference, University of Cambridge</i>   |                             |

## Graduate Research Studentship

*Toshiba Research*

2021

GBP 150k

## Cum Laude

*University of California, Santa Cruz*

2015

## INVITED TALKS

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### Panelist with Sergey Levine, Animesh Garg, and Dhruv Shah

*Conference on Robot Learning – RemembeRL Workshop*

2025

### Understanding and Benchmarking Memory in RL

*Conference on Robot Learning – RemembeRL Workshop*

2025

### Deep Decision Making

*University of Macau STGC 8003 Guest Lecture*

2025

### Deep Reinforcement Learning with Memory

*Emotional AI Forum at the University of Macau*

2025

### Reinforcement Learning with Memory

*University of Cambridge R271 Guest Lecture*

2024

### Value-Based Reinforcement Learning

*Cambridge ELLIS Unit Summer School on Probabilistic Machine Learning*

2024

### Reinforcement Learning under Partial Observability

*University of Macau Research Talk*

2024

### An Introduction to Reinforcement Learning

*Toshiba Cambridge Research Seminar*

2023

## TEACHING EXPERIENCE

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### Instructor (Macau)

- Mean student evaluation score over all modules: 5.77/6.00 (80.3% response rate)
  - Real-time captioning and translation used to support non-native English speakers
- Coordinating marking with teaching assistants
- Modules designed and taught
  - Special Topics in AI: Decision Making and Reinforcement Learning (2025, 2026; 53, 74 students)
  - Introduction to Deep Learning, (2024, 2025; 74, 110 students)

### Instructor (Cambridge)

- Modules designed and taught
  - Advanced Topics: Deep Reinforcement Learning, (2024; ~30 students)

### Teaching Assistant (Cambridge)

- Programming in C/C++ (2023; ~10 students)
- Introduction to Robotics (2022; ~40 students)
- Mobile Robot Systems (2021; ~40 students)

### Thesis Supervision

- Yutao Chen, PhD, **Deep Mixture Modeling with Natural Gradients**, Macau (2025–)
- Xiaoming Liu, PhD, **Offline Reinforcement Learning for Mobile Manipulation**, Macau (2025–)
- Wang Zekang, MS, **Evaluating and Rethinking Memory in RL**, Macau (2024–2026)
- He Zhe, MS, **Efficient Memory Diagnostic Benchmark and Parallelized RL**, Macau (2024–2026)
- Dulhan Jayalath, MPhil, **Reconstructing Markov States in Decentralised Multi-Agent RL**, Cambridge (2022–2023)
- Mark Li, Part II, **Intrinsically Go-Explore in Reinforcement Learning**, Cambridge (2023–2024)
- James Read, Part II, **Offworld SLAM**, Cambridge (2021–2022)

## SKILLS

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**Core Competencies**

Deep Learning, Reinforcement Learning, Sequence Modeling, Multi-Agent Reinforcement Learning, Robotics

**Programming Languages**

Python

**Models**

World Models, RNNs, Transformers, Generative Models, Variational Models

**Software**

SolidWorks, Linux, ROS2, MATLAB, Git

**ML Frameworks**

JAX, PyTorch, Numpy, Equinox

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**SERVICE AND OUTREACH**

Director, AI and Robotics Summer School, University of Macau (2026)  
Session Chair, IEEE Big Data Conference, Macau (2025)  
Faculty Affiliate - Moon Chun Memorial College, University of Macau (2025-2026)  
PhD Defense Committee - Sun Binbin, University of Macau (2025-2026)  
MS Thesis Examination Chair - Huanrong Liu, University of Macau (2026)  
MS Thesis Examination Chair - Ting Huang, University of Macau (2026)  
MS Thesis Examination Chair - Zhijia Tao, University of Macau (2026)  
MS Thesis Examination Chair - Leandro Schlosser Acosta, University of Macau (2025)  
MS Project Examiner - Artificial Intelligence, University of Macau (2025, 2026)  
BS Thesis Examiner - Artificial Intelligence, University of Macau (2026)  
Mentor, UC Santa Cruz (2023)  
Volunteer, Boys and Girls Club, Tucson (2018)

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**PEER REVIEW**

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|------------------|-----------------------|---------------------|
| ICLR (2024-2026) | ICML (2024-2026)      | NeurIPS (2024-2026) |
| AAAI (2025)      | ICRA/RA-L (2020-2021) | RSS (2024)          |
| CoRL (2026)      |                       |                     |